SRS Template: Student Guide

Guide for students · PGCP VLSI students' project · Design, Verification & FPGA Validation

What this document is

The System Requirements Specification (SRS) states **what your design must do, how well it must do it, and how you will prove it**. It turns your Week 1 problem statement and objectives into requirements that can be tested. It is a Week 3 deliverable. The performance, power and area (PPA) targets you set here are what your Month 3 results will be judged against, so they need real numbers.

The SRS says nothing about *how* you will build the design. That belongs in your HLD and LLD.

How to use the template

- Fill in the yellow cells. The grey lines under each heading explain what to write, and you can delete them when you finish.
- Write each requirement as one numbered, testable statement, such as "The system shall ...".
- Give every requirement an ID (FR-01, PERF-01, PWR-01, AREA-01, FPGA-01 and so on). Section 10 must link every ID to how you will check it.
- Fill in the document control table each time you change the document.

What each section contains

Document control and header: project title, group, members, Group Mentor, version and date, plus a table recording each change.

1. Introduction

- 1.1 **Purpose:** one or two sentences on what the document specifies and who it is for.
- 1.2 **Scope:** what the design does and does not do, and what is in scope for verification. Take this from your Week 1 scope statement.
- 1.3 **Definitions and acronyms.**

- 1.4 **References:** papers, standards and datasheets, including your Week 2 survey sources.

2. Overall description

- 2.1 **System context:** a diagram showing your design as one block, with everything it connects to outside it.
- 2.2 **Design platform and tools:** ASIC or FPGA, technology or board, design language, verification methodology and tool flow.
- 2.3 **Assumptions and constraints.**

3. Functional requirements: what the design must do, including features, operating modes, data formats, error handling and reset behaviour. Each row has an ID, the requirement, a priority (Must, Should or Could) and how it will be verified.

4. Performance requirements: clock frequency, throughput and latency. Give a target and a minimum acceptable value, with the conditions (process corner, voltage, temperature, workload).

5. Power requirements: dynamic, leakage and total power, with a budget at a stated frequency and voltage. Note any low-power modes you plan, such as clock gating or sleep states.

6. Area requirements: for ASIC, gate count or mm^2 and memory sizes. For FPGA, LUTs, flip-flops, BRAM and DSP blocks. If your targets conflict, state which of power, performance and area takes priority.

7. Interface requirements

- 7.1 **Interface summary:** each interface, its protocol (AXI4-Lite, APB, SPI, UART or custom), direction and what it connects to.
- 7.2 **Signal list:** name, direction, width, clock domain and description for every external signal.
- 7.3 **Clocks and resets:** frequency, type, and whether reset is synchronous or asynchronous.
- 7.4 **Timing and handshake rules:** handshake protocol, latency, and setup/hold expectations at the boundary. A timing diagram helps.
- 7.5 **Register map:** only if software accesses the block.

8. Verification requirements

- 8.1 **Verification strategy:** levels (block and integration) and approach (directed, constrained-random, formal).

- 8.2 **Testbench environment:** driver, monitor, scoreboard or reference model, assertions and coverage collector.
- 8.3 **Coverage goals:** a target for statement, branch, toggle, FSM, functional and assertion coverage, with any exclusions explained.
- 8.4 **Test plan:** one row per test, linked to the requirement it checks, with pass criteria.
- 8.5 **Exit criteria:** what must be true to call verification complete, for example all tests pass, coverage targets met, no open critical bugs, lint and CDC clean.

9. FPGA validation requirements

- 9.1 **Scope and objectives:** what you will validate on hardware, especially what simulation can't fully check.
- 9.2 **Platform and tools:** board and part number, toolchain, board clock, and debug method (ILA, logic analyser, UART or host software).
- 9.3 **Implementation targets:** clock frequency on board, timing closure (worst negative slack of zero or better), and LUT, flip-flop and BRAM/DSP utilization.
- 9.4 **I/O and pin mapping.**
- 9.5 **Validation test plan:** procedure on the board, expected result and observed result for each test.
- 9.6 **Correlation with simulation:** how you compare hardware results with simulation and handle mismatches.
- 9.7 **Exit criteria.**
- If your group won't prototype on FPGA, write "Not applicable" in 9.1 with the reason and delete the rest of the section.

10. Traceability and acceptance: one row per requirement ID from sections 3 to 9, showing the method (simulation, assertion, formal, coverage, synthesis, STA, FPGA or review), the test or report that provides the evidence, and the status.

11. Approval: signature lines for your group and your Group Mentor.

An example of a testable PPA entry

The values below are placeholders. Your own numbers come from your survey and objectives.

ID	Parameter	Target	Minimum	Conditions	Verified by
PERF-01	Clock frequency	X MHz	Y MHz	typical corner, nominal voltage	post-synthesis STA
PWR-01	Total power	≤ X mW	≤ Y mW	at X MHz, nominal voltage	power analysis
AREA-01	Area	≤ X mm²	≤ Y mm²	after placement	synthesis or P&R report

How to choose your numbers

- Tie your targets to your Week 2 survey, so they can be defended against existing work. For example, "within X% of the best published power".
- Make every number testable. "Low power" can't be checked, but "≤ X mW at Y MHz" can.
- State the conditions with every target, so the same measurement can be repeated.

Common mistakes to avoid

- Vague words like "fast" or "efficient" with no number.
- Targets with no stated conditions.
- Interfaces described in a paragraph with no signal-level detail.
- Design decisions written as requirements, such as "use a 5-stage pipeline", instead of the need behind them.
- Requirement IDs that never appear in the traceability table.

Before you submit

- Share your draft with your Group Mentor before the deadline of [date].
- Keep your Week 2 survey open while setting targets.
- Update the SRS if your design changes, and record the change in the document control table.